Technical report: RöstiBridge

- **Track:** Track 2A, UZH: Detecting Cross-Lingual Semantic Differences in Swiss Government Websites
- **Event:** Online
- **Team:** RedPandas: Christopher Yazigi, Giovanni Yazbeck, Alfredo Pio Yazbeck
- **Demo:** Firefox extension in `src/extension` (`dist/roestibridge-0.1.0.xpi`); screenshots in `docs/`

1. Summary

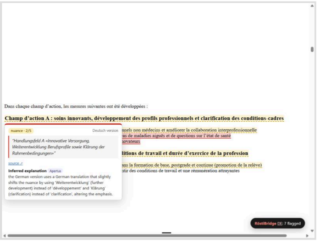
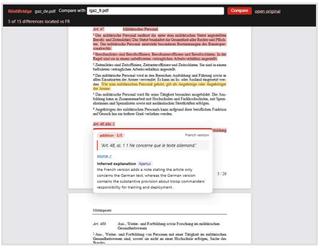
Swiss authorities publish the same page in up to four languages, and the versions do not always say the same thing: sentences are dropped, numbers or deadlines change, or “must” becomes “may”. RöstiBridge brings token-level recognition of semantic differences (RSD) to the reader. It is a Firefox extension that highlights passages of an official page (federal, cantonal, RTS news, or a parliamentary PDF) whose meaning may differ from the reference language version. Hovering a highlight shows the other version’s excerpt as a citation and an Apertus explanation.

The same single Apertus prompt is benchmarked on SwissGov-RSD [1]. Apertus-v1.5-70B reports difference spans, which are projected onto tokens; 3-sample self-consistency averages them. On a 120-document dev subset this gives a macro token-level Spearman of **0.131**, against 0.083 for a model-free number-mismatch baseline and 0.096 for the raw single-sample spans. Test-split predictions (168 documents) are in `predictions/test_70b_n3/`.

2. Architecture

```
page / PDF in language X ──► extract blocks ──► find counterpart in language Y
  (content script / pdf.js)                      hreflang · switcher · /de/↔/fr/ URL
                                                 · Apertus path matching (admin.ch CMS)
                                                 · OpenParlData affair siblings (parlament.ch PDFs)
                                                 · Bing-News RSS + Apertus story matching (RTS→SRF)
              │                                             │
              └──────► background.js: chunk aligned blocks (≈3.5k chars, proportional
                       window ±4 blocks) ──► Apertus (shared/prompt.json) ──► JSON
                       {a, b, type, severity, explanation} ──► cache (storage.local)
              ◄────── highlight `a` excerpts (whitespace-agnostic matching across nodes /
                       text-layer spans) + hover card (citation `b` + explanation)
```

- **Federal vs. cantonal logic** (`shared/sites.js`):
    - Federal domains are compared with FR and DE (whichever the page is not in).
    - Bilingual cantons (BE/FR/VS/GR) are compared with the other official language.
    - Monolingual cantons are compared with their official language.
- **Key handling:** the API key lives in `browser.storage.local` and is only read by the background script. Page scripts never receive it.
- **Benchmark pipeline** (`src/roestibridge`): a stdlib-only Python client (`llm.py`), span → token projection (`rsd.py`), the metric (`metrics.py`, a faithful re-implementation of the official `evaluate_predictions_admin.py` pooling) and a CLI. It reads the **same** `prompt.json` as the extension, and a unit test enforces this.

Federal page: hover card	Parliamentary PDF: hover card
 Dans chaque champ d’action, les mesures suivantes ont été développées : **Champ d’action A : soins innovants, développement des profils professionnels et clarification des conditions-cadres** nnels non médecins et améliorer la collaboration interprofessionnelle as de maladies aiguës et de questions sur l’état de santé novateurs **litions de travail et durée d’exercice de la profession** ans la formation de base, postgrade et continue (promotion de la relève) atir des conditions de travail et une rémunération attrayantes *“Handlungsfeld A «Innovative Versorgung, Weiterentwicklung Berufsprofile sowie Klärung der Rahmenbedingungen»”* **Inferred explanation** the German version uses a German translation that slightly shifts the nuance by using 'Weiterentwicklung' (further development) instead of 'développement' and 'Klärung' (clarification) instead of 'clarification', altering the emphasis. 	 RöstiBridge gaz_de.pdf Compare with /gaz_fr.pdf Compare open original 5 of 15 differences located vs FR *Art. 47* Militärisches Personal ¹ Das militärische Personal umfasst die unter dem militärischen Statut angestellten Berufs- und Zeitmilitärs. Das Statut beinhaltet die Gesamtheit aller Rechte und Pflichten. Das militärische Personal untersteht besonderen Bestimmungen des Bundespersonalrechts. ² Berufsmilitärs sind Berufsoffiziere, Berufsunteroffiziere und Berufssoldaten. In der Regel sind sie in einem unbefristeten vertraglichen Arbeitsverhältnis angestellt. ³ Zeitmilitärs sind Zeitoffiziere, Zeitunteroffiziere und Zeitsoldaten. Sie sind in einem befristeten vertraglichen Arbeitsverhältnis angestellt. ⁴ Das militärische Personal wird in den Bereichen Ausbildung und Führung sowie in allen Einsatzarten der Armee verwendet. Es kann im In- oder Ausland eingesetzt werden. Wer zum militärischen Personal gehört, gilt als Angehörige oder Angehöriger der Armee. ⁵ Das militärische Personal wird für seine Tätigkeit besonders ausgebildet. Die Ausbildung kann in Zusammenarbeit mit Hochschulen und Fachhochschulen, mit Spezialistinnen und Spezialisten sowie mit ausländischen Streitkräften erfolgen. ⁶ Angehörigen des militärischen Personals kann aufgrund ihrer beruflichen Funktion auf Gesuch hin ein tieferer Grad verliehen werden. *Art. 48 Abs. 1* bildung *“Art. 48, al. 1 1 Ne concerne que le texte allemand.”* **Inferred explanation** the French version adds a note stating the article only concerns the German text, whereas the German version contains the substantive provision about troop commanders' responsibility for training and deployment. 5 / 20 Militärgesetz *Art. 48b* Aus-, Weiter- und Fortbildung sowie Forschung im militärischen Gesundheitswesen ¹ Aus-, Weiter- und Fortbildung von Personen mit einer Tätigkeit im militärischen Gesundheitswesen sind, soweit sie nicht an einer Hochschule erfolgen, Sache des Bundes

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` (CSCS-hosted, OpenAI-compatible endpoint). `Apertus-v1.5-8B` can be selected in the options for on-prem use.
- **How it is used:** prompting-only inference with JSON output (`response_format: json_object`, with a fallback when the endpoint does not support it). Apertus is used for:
    1. difference detection with severity and explanation (core RSD);
    2. matching translated URL slugs on the new admin.ch CMS;
    3. turning an RTS headline into German search queries and deciding whether an SRF article is the same story;
    4. picking the translated sibling of a parliamentary document.
- **Prompt:** `src/extension/shared/prompt.json` (`rb-prompt-v1`). The system prompt defines what counts as a difference (omission/addition, numbers/dates, obligation/scope, legal terms, actionable nuance) and what does not (grammar, word order, free translation). It asks for exact, short excerpts.
- **Benchmark settings:** temperature 0 for the first sample and 0.7 for samples 2–3, `max_tokens` 2000, one call per document pair. Token usage (from the API) is about 0.82M prompt and 0.25M completion tokens for the 3-sample test run (`raw_*.jsonl` → `usage`).
- **No other model is used** anywhere in the pipeline or for evaluation.

4. Data

- **SwissGov-RSD** [1] (`ZurichNLP/SwissGov-RSD` on Hugging Face) is downloaded at run time into `data/swissgov-rsd/` and is not committed.
- Development used a deterministic, evenly spaced **40-document subset of the dev split per language** (`--limit 40`, 120 pairs) to stay within the API budget.
- **The test split was never evaluated or inspected:** `evaluate --split test` exits with an error.
- Test predictions were produced once with the prompt and post-processing frozen on dev.
- The extension processes public government pages on the fly and stores nothing beyond a local cache of model answers.
- Note: the dataset's DE/FR/IT texts contain U+FFFD replacement characters in place of accented letters (e.g. `L�approvisionnement`). The prompt tells the model to copy excerpts verbatim "even if characters look garbled", and the projection treats `�` as a word character.

5. Evaluation

Metric: the official one. For each language pair, all tokens of side A and side B with gold ≠ −1 are pooled (predictions are padded or truncated to the gold length), global Spearman is computed, and the macro average is taken over EN–DE/FR/IT. Our implementation matches `scipy.stats.spearmanr` on the pooled vectors (checked in development).

Setup (dev subset, 40 docs/pair)	DE	FR	IT	macro
Number-mismatch baseline (no LLM)	0.102	0.068	0.080	0.083
Apertus-70B, 1 sample, all spans at severity/5	0.136	0.056	0.096	0.096
+ drop severity 1, omission/addition = 1.0	0.157	0.069	0.100	0.109
+ number-mismatch prior (max)	0.159	0.074	0.104	0.112
+ 3 samples, averaged (submitted)	**0.191**	**0.077**	**0.125**	**0.131**

Ablations (no extra API calls; `python -m roestibridge.ablate`):

Variant	Macro	Result
Binary span scores instead of severity	0.095	Worse
Damping long spans by length	±0.001	No gain
Paragraph-context smoothing	0.099	No gain

Error analysis (FR, 1 sample): - **Low precision.** Of the tokens we mark, 226 are true positives and 1,752 are false positives. Apertus over-reports “different wording, same meaning” spans at low severity, which motivated the severity ≥ 2 cut. Recall is limited too (1,119 gold-positive tokens missed): many gold differences are small structural omissions such as link labels and list items. - **Span projection is reliable:** 376 of 382 returned excerpts were located in the token sequence. - **FR is the weakest pair.** The FR documents are longer (461 vs 363 tokens on average for DE) and the FR texts carry the most `�` corruption.

Extension checks. These ran end to end in Firefox 158 (headless, WebDriver BiDi via puppeteer-core) on the packaged extension against live sites on 16 Oct 2026. Apertus-70B was used throughout. The counts vary between runs because the model is not deterministic.

Case	Result
bfe.admin.ch/fr → DE	3 highlighted, e.g. a paragraph on the IAEA General Conference that is missing in the German version
be.ch/fr → DE (bilingual canton)	1 highlighted, a tone shift (“beau palmarès” vs “brillieren”)
bag.admin.ch FR → DE	Counterpart found via Apertus slug matching (`agenda-soins-de-base` → `agenda-grundversorgung`); 0–9 differences depending on the run, mostly list items
ch.ch/it → DE and FR (federal: both)	0 differences on the homepage
RTS Amaudruz article → SRF	SRF story found and linked (“SVP-Nationalrätin Céline Amaudruz will Bundesrätin werden”); 8 differences highlighted
parlament.ch PDF link (DE)	Redirected into the RöstiBridge viewer; FR counterpart found via OpenParlData affair 234959 (`Kinderabzug = 12'000 Franken` → `déduction pour enfants = 12 000 francs`); the tables agree (0 differences)
Fedlex consultation PDF (DE↔FR)	16 differences, including articles where the French reads “Ne concerne que le texte allemand” while the German changes the provision; 5 located on the text layer

6. Limitations

- **Absolute scores are low.** This is consistent with [1], where prompted LLMs struggle on document-level cross-lingual RSD. The dev subset is small (120 pairs), so the post-processing choices may be slightly overfit.
- **Extension chunk alignment is proportional (± 4 blocks).** When a page has very different structures across languages, content outside the window can be reported as “missing”. Apertus sometimes quotes the wrong side; such spans are simply not highlighted (10 of 15 in the PDF case).
- **Counterpart discovery depends on site conventions.**
    - The admin.ch slug matcher needs both homepages to list the page.
    - RTS→SRF depends on a third-party news search (Bing News RSS, DuckDuckGo fallback), and SRF is independent journalism, not a translation. Differences there mean coverage differences, not translation errors.
- **Language metadata:** OpenParlData labels some French parlament.ch documents as `it`. We therefore ask Apertus to pick by title rather than trusting the language field.
- **Packaging:** the extension is unsigned (temporary add-on) and uses Manifest V2 for blocking `webRequest` (PDF interception).

- **No human evaluation of the extension’s highlights** was done in the time available.

7. Reproducibility

- `make run` (Docker, no GPU) runs four steps:
    1. unit tests;
    2. re-scoring of the shipped raw Apertus outputs for dev (exactly the numbers above, no API needed);
    3. the model-free baseline;
    4. rebuilding the submitted test predictions from the shipped raw outputs.

  With `LLM_API_KEY` set it also runs a live Apertus prediction and evaluation on `LIVE_LIMIT` dev documents per language.
- `make predict-test` regenerates the test predictions (168 × 3 calls).
- Apertus output at temperature 0 is not perfectly deterministic on the hosted endpoint. That is why the raw responses are shipped and all scores are recomputed from them.

8. Next steps

- **Sentence alignment before prompting.** Align with the LLM or embeddings, then ask per aligned pair. This should improve both recall and the “missing” false positives.
- **Calibrate severities per language** on dev.
- **Try 8B with more samples** for an on-prem budget.
- **Ship a signed MV3 build:** declarativeNetRequest for PDFs, plus a “report this discrepancy to the authority” button that drafts a correction request with the citation.

License

Creative Commons Attribution 4.0 (CC-BY-4.0). pdf.js (Apache-2.0) is bundled in `src/extension/pdf/lib/` with its licence.

References

[1] Wastl et al. *SwissGov-RSD: A Human-annotated, Cross-lingual Benchmark for Token-level Recognition of Semantic Differences Between Related Documents.* arXiv:2512.07538 / ACL 2026. https://arxiv.org/abs/2512.07538